

Introducing weave-cli

Powering the Next Generation of AI Agents

— github.com/maximilien/weave-cli —

Your Universal VDB Development Tool

What Are Vector Databases?

Specialized databases optimized for storing and querying high-dimensional vector embeddings that represent semantic meaning of data.

Why Critical for AI Agents?

- Enable semantic search beyond keyword matching
- Power RAG systems with contextual knowledge retrieval
- Support real-time similarity search at scale
- Essential for LLM memory and context management

Market Growth

\$2.2B

2024 Market Size

\$10.6B

2032 Projection

23.5% CAGR

How They Work: Embeddings & Vector Space

The Process

1. Embedding Generation

Convert text, images, audio to numerical vectors using ML models

2. Vector Indexing

Organize vectors using algorithms like HNSW, IVF for fast search

3. Similarity Search

Find nearest neighbors using cosine similarity or Euclidean distance

Key Concepts

High-Dimensional Space

Vectors typically have 768-4096 dimensions, representing complex semantic relationships.

Semantic Proximity

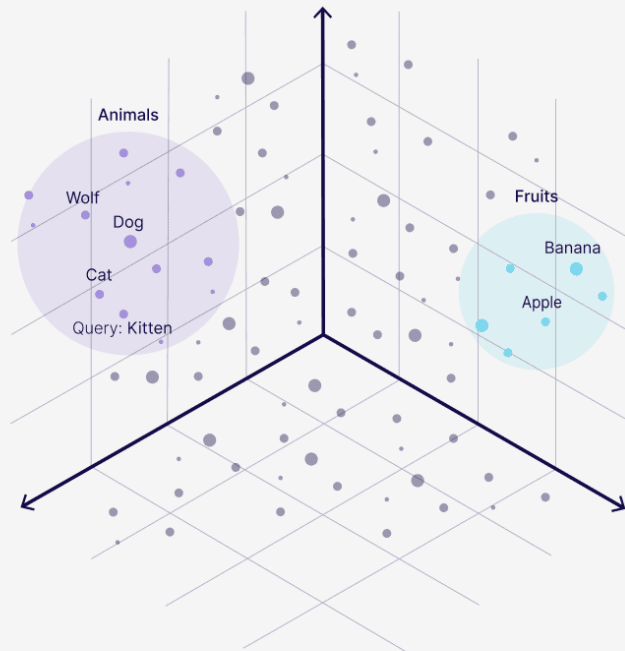
Similar concepts cluster together in vector space, enabling contextual search.

ANN Algorithms

Approximate Nearest Neighbor algorithms trade perfect accuracy for millisecond query times.

How They Work: Embeddings & Vector Space (cont.)

Vector search returns similar items **based on their semantic meaning**, rather than exact term matches.



Credit: weaviate.io/blog/vector-search-explained

The Hybrid Search Solution

The Challenge

Pure semantic search can miss exact matches that traditional keyword search excels at finding.

1 Semantic Search

- Understands context and intent
- Finds conceptually similar results
- Works across languages

2 Keyword Search

- Precise exact matches
- Fast for specific terms
- Good for technical queries

Hybrid Search = Best of Both Worlds

Combine vector similarity with keyword matching using RRF for optimal accuracy.

Data Processing: The Foundation

Chunking Strategy Matters

Optimal Size

256-512 tokens balance

Overlap

10-20% prevents loss

Metadata

Enrich with tags

Multimodal Processing



Documents

PDF parsing, OCR extraction



Images

CLIP, vision transformers



Audio

Whisper transcription



Video

Frame sampling, fusion

Pro Tip

Quality processing = Better retrieval.
Invest in robust pipelines.

One Tool for All Vector Databases

The Problem

Each VDB has unique APIs and workflows. Switching means learning different interfaces.

weave-cli Solution

✓ Unified Interface

One CLI for all VDBs

✓ Fast Prototyping

Test multiple VDBs

✓ Dev to Prod

Seamless transitions

✓ AI-Powered

Agent-assisted ops

✓ Benchmarking

Compare & optimize

✓ Open Source

MIT License

Use Cases

- Multi-VDB evaluation
- Provider migration
- RAG development
- Optimizing search (multi search)
- Benchmarking

Integration Time

~10x Faster

Live Demo Time!



★ Star us on GitHub

github.com/maximilien/weave-cli

Fast Setup

5 min to first query

Multi-VDB

10+ VDBs

Open Source

MIT License